advantage estimation [Schulman 16]; Section 6.2.5 gives the exact objective. GPU-parallel simulation supplies the required throughput [Makoviychuk 21, Rudin 22], which is why compact actuator surrogates replace the full hydraulic plant during training.

2.7.2 Structured Priors for Policy Learning

Learning from scratch would discard reliable kinematics. Residual RL instead adds a bounded learned correction to a controller at runtime [Johannink 19, Silver 18]. Policy distillation moves the prior to training: a privileged DLS teacher supplies demonstrations then the student acts alone [Ross 11]. This thesis distils the teacher and then anchors PPO updates to the frozen student (Section 6.4). The deployed actor remains the sole controller.

The idea parallels real-world fine-tuning that retains a simulation-trained value function [Yin 25], but here the retained object is the reference policy. Anchoring was introduced because unconstrained PPO improved reward while degrading the physical tracking criterion (Section 6.3).

2.7.3 Reinforcement Learning on Hydraulic Machines

Deep RL has reached full-size hydraulic excavators in simulation-to-machine tracking [Egli 20] and general automation [Egli 22]. Spinelli et al. [Spinelli 24] extended it to commercial material handlers with underactuated tools. Maitan et al. [Maitan 26] learned force control on a physical cylinder, using a contraction-metric filter to bound exploration. These studies establish scale and feasibility, but do not combine a randomized plant family, online actuator adaptation, a privileged kinematic teacher and a matched comparison of actuator representations.

A policy can also compensate for a poor actuator model. Chapter 7 therefore reports rollout error, command magnitude and smoothness, and cylinder-level residuals alongside task tracking. Equal task error alone is not evidence of equal model quality.

2.8 Research Gap and Thesis Positioning

The literature leaves a specific combination unresolved. Hydraulic control provides structure but requires maintained plant models; learned dynamics replace derivation with data but depend on identifiable excitation; actuator surrogates and latent adaptation are established mainly for electric legged robots; and hydraulic RL studies do not isolate actuator representation under a matched protocol. Table 2.1 summarizes this position.

The five criteria are: (I) learned hydraulic actuation, (II) data audited for causal input identifiability, (III) online latent adaptation, (IV) a structured controller retained as a bounded prior, and (V) a matched actuator comparison. Parenthesized marks denote partial or implicit coverage.